1
0:0:0-->0:0:16
Hello, everyone. And welcome to cs 584 otherwise known as natural language processing. So my name is houston crow. I'll be your professor for this semester. And i'm here too, today at least

2
0:0:16-->0:0:32
discuss what this course is about, what nlp is about and give an overview of the course structure. In particular, what we're going to go through every video lecture, how the topics will relate to one another. It's basically the syllabus and how the

3
0:0:32-->0:0:46
topics in the syllabus and ah modules in the syllabus relate to one another. And ideally, how you can use the topics in one to you know learn more about the others and so forth, how one builds on on each. Although as you'll see, some of them are a bit staggered. But

4
0:0:47-->0:1:3
for today, I want to do that as well as give you a general overview as to like what we want to accomplish this semester in terms of learning nlp in other words, what do we want to get out of this course the course goals?

5
0:1:7-->0:1:22
So i'm going to start off by it's always hard to start off a natural language processing course, because i'm using language to communicate this to you. And there's always like the tendency to get really meta when it comes to natural language processing,

6
0:1:22-->0:1:33
or a topic that's about one of the tools you use to communicate the topic will be discussing things like this throughout the semester.

7
0:1:33-->0:1:46
One way we can begin is simply by writing out and defining what is natural language processing,

8
0:1:48-->0:2:13
otherwise known as nli hope you can see this this whiteboard. One way to begin defining this, defining natural language processing is by first understanding what we mean by each of these constituent parts. What is

9
0:2:13-->0:2:32
something that's natural? What is a natural language? And what does it mean to process? It so much so that we can say we are doing natural language processing. you knowSo what is a language? Well,

10
0:2:33-->0:2:41
a language is basically a tool I use to communicate or we use to communicate with other

people or things.

11
0:2:41-->0:2:49
So right now i'm using a language, i'm using english to communicate, concepts, whatever concepts my

12
0:2:49-->0:3:6
uh you know brain can kind of like instantiate, and whatever concepts I i'm pretty sure you'll understand. That's what I use words to communicate. It's a bit weird thinking about this stuff as i'm using that as i'm using language. But

13
0:3:7-->0:3:30
it's, for the sake, ofthis course we're going to consider language simple simple text. Our language, broadly speaking, is just a communication tool,

14
0:3:34-->0:3:40
which will be text based for this class, more on this in a sec.

15
0:3:40-->0:4:2
So in other words, there's tons of other tools I can use that are not text based. For example, i'm talking to you using regular audio words. I'm not writing these words, as i'm saying them,

16
0:4:2-->0:4:19
right? But that communication is not, as a result, text based. I could also knock on things or make make sounds. umCan whistle. I don't want to do it right now, but um I could use my hand. I can use sign language you know

17
0:4:19-->0:4:38
and and tell you things with my with my hands that is also communication. And that's non textual. Nop broadly speaking, does capture that. However, just for this class, we're going to focus on text based communication. So as we'll see in a later class, this is really just

18
0:4:38-->0:4:48
referring to finite strings

19
0:4:50-->0:4:51
of symbols.

20
0:4:57-->0:5:20
wellNow, what does it mean for things to be natural natural for the most part? Means it's occurring, oops, occurring in nature, quote, unquote, okay wellwhat is nature?

21
0:5:20-->0:5:36

What's basically anything we, as humans have? Not like poor like operated on to to produce that or something that is natural or something that we have not we, as humans have not processed mechanically, if that makes sense.

22
0:5:36-->0:5:37
So in other words,

23
0:5:37-->0:5:55
uha natural organism is one that simply occurred, right? From the ground or from the sky or from the rest of what we would call nature. And we haven't done anything to it. So let's say a plant a plant is a natural thing. This is all for the sake of

24
0:5:55-->0:6:12
today's class. I want to get to philosophical, but a plant is a natural thing. If we as humans have not really how do I say like consciously dissected it and done some process to it and transformed it.

25
0:6:12-->0:6:28
Now, of courselanguage transforms every day, and we are the ones transforming it for the most part. So I know it's hard to argue. I mean you can always argue oh language is almost never natural, but hear me out.

26
0:6:28-->0:6:44
There are things, as we'll see in another lecture, called formal languages, which are far from natural and very much constructed by us humans. Formal languages include things like mathematics. Um

27
0:6:44-->0:6:58
Python is a natural language, our coding languages, uhother formal languages may or may not include music that could also be considered a natural language like musical notes. Another formal language might be the language of logic.

28
0:6:58-->0:7:0
So, for example,

29
0:7:1-->0:7:28
a formal, so this is natural, as opposed to a formal a formal language, would be math. It could be a logical theory like set theory, other types of logic, coding languages like c plus plus.

30
0:7:29-->0:7:54
andThere's a variety of other ones. Natural. On the other hand, a natural language basically just refers to what we, as humans speak for this class. That's what that means. And so a natural language is just

31

0:7:54-->0:8:10
a communication tool, welike we humans speak or use, in this case, a text based one. So as you probably would have guessed by now, we're just talking about like words words, sentences, paragraphs.

32
0:8:11-->0:8:46
Now, for the final part of this processing, now, at the end of the day, processing just refers to interpretation or evaluation of meaning. All good with that. Processing is just

33
0:8:46-->0:8:50
the interpretation or evaluation of the meaning of something.

34
0:8:50-->0:8:58
Now, again, I don't want to get too philosophical, but when we say meaning, we really just mean some

35
0:8:58-->0:9:16
well one way to say when people say meaning oftentimes they mean interpretation or what does interpretation mean? Meaning? okayWhat do both of these things mean? Meaning usually refers to some other longer. We might call set of words or sentence

36
0:9:17-->0:9:35
or idea. Right? Some other physical or non physical or abstract thing concept. Or it could even be another set of words, right? That a certain other set of words insinuate,

37
0:9:36-->0:9:50
sofor example, of that words have meanings, and we often call those type those those meanings, definitions. But what is a definition? Right? If you think about it a definition, which is the meaning of a certain word,

38
0:9:50-->0:9:57
a meaning, not v meaning because there are many definitions of words, aa definition.

39
0:9:57-->0:9:59
If you just look at it, plain and simple,

40
0:9:59-->0:10:1
is really just another set of words.

41
0:10:1-->0:10:23
For example, if I were to define the word run, if I were to process this myself as a human being, the word run, okay wellI would just look at the definition. What is the definition?

42
0:10:23-->0:10:59

it'sA and at first, usually you'd see the fact that it's a verb. It would be, if I would say it's the act of using one's legs to speedily travel by

43
0:11:0-->0:11:33
walking in longer impulse strides. So what i'm saying here, a definition, I don't know if this is the exact definition, but it's a definition, right? Of run. The verb to run is basically if you walk,

44
0:11:33-->0:11:48
but with in every stride, make it longer and impulse, right? In other words, you could walk really fast with long strides. But if they're just continuous strides, you'd still be walking. What makes you run is if you bounce or impulse,

45
0:11:48-->0:11:59
in like your feet, every stride or bounce off of every stride that then you could either be skipping or effectively running if you do that frequently.

46
0:11:59-->0:12:3
In other words, you could think of it as like skipping skipping

47
0:12:3-->0:12:19
very fast and often with your legs is a base is basically running. Anyways, all i've done, though, to interpret this word run is associated with a bunch of other words, or the corresponding concepts.

48
0:12:21-->0:12:42
so whenI say meaning when we say meaning or interpretation, really, what we mean is another concept or another string of words whose interpretation we already have. Hopefully that makes sense. okaySo that's what processing is, right? It's the act of

49
0:12:44-->0:13:5
mapping these words, to other words and concepts. So what does like putting it all together? What does natural language processing really mean? Well, as humans, we already do this, right? Otherwise, you knowhow are you understanding what i'm saying? And

50
0:13:5-->0:13:22
how am I understanding what i'm saying? If i'm not processing my own natural language? So when we say natural language processing, we don't mean trivial you know us hearing each other. We're not talking about humans. We're talking about how machines or computers

51
0:13:22-->0:13:44
process natural language. So putting this all together, natural language processing, i'll erase this. And there's a slightly deeper definition, which i'll explain in a sec.

52
0:13:44-->0:13:54

soIf we put all these things together, right? Nlp is basically

53
0:13:58-->0:14:23
as a field methods. Oftentimes, what we'll call formal methods. Computers use to interpret natural language,

54
0:14:24-->0:14:46
or in our case text. soThis class is largely going to be about techniques or methods. Again, i'll explain what formal means in a separate, in our next one or two lectures. Formal methods, computers used to interpret natural language.

55
0:14:47-->0:14:55
Hopefully that makes sense. Basically, we're going to be figuring out how a computer can understand what we're saying.

56
0:14:58-->0:14:59
So

57
0:14:59-->0:15:16
This course at its core is going to refer to and go over the key concepts we need to understand and kind of grasp in order to get a computer or basically use formal language, right? Because computers,

58
0:15:16-->0:15:28
as you might have guessed, use formal languages to to do things.

59
0:15:28-->0:15:43
So really, if you think about it, nlp is really about using formal language

60
0:15:48-->0:16:12
and methods to process natural language. Sorry, if my handwriting is a bit off, as you'll see, we'll use python, which is a formal language to try to understand english, which is a natural language.

61
0:16:13-->0:16:31
You see how I can get a little confusing or meta. But that's what this is about. And at the end of the day, while there is some theory behind this, you know umwhich, as we'll see, has to do with the way we as humans formalize our natural language into things like

62
0:16:31-->0:16:36
words and grammar sentences and so forth and morphology.

63
0:16:36-->0:16:47
A lot of it is just techniques that we've seen work often ranging from you know using formal

grammar is to crystallizing formalized language

64
0:16:47-->0:17:3
and know how to properly interpret it to neural networks you know that rely heavily or almost entirely on statistics is just capture general patterns on interpretations. Whatever it takes at the end of the day. These are the techniques

65
0:17:4-->0:17:23
people have used to get computers and formal systems to interpret natural language that we use. So basically, we're trying to get machines to understand humans for. Now, just the text okay moving on.

66
0:17:24-->0:17:51
I'm going to close this word. Give me one moment. okayNext, i'm going to go over the syllabus. So hopefully you can all see what's going on here. And in fact,

67
0:17:52-->0:18:13
I will I wonder if I can, there we go. So the first part of this course is just today an intro to nlp. umThere is a python knowledge check or quiz,

68
0:18:13-->0:18:26
as well as a general coding assignment do. And uh in python, a formal language. I'm going to assume we all have an idea of python or the basics of it.

69
0:18:26-->0:18:32
Next, we'll talk about machine learning basics, because we

70
0:18:32-->0:18:49
we'll need to know machine learning in order to do some of these important techniques. We all know that language, natural language is changing all the time based on how many people use. you knowFor example, what words to mean? What? Right?

71
0:18:49-->0:19:4
soInherently, there's a bunch of randomness in our and the way we use language. And as a result, language as a whole that randomness we have to use statistics in order to capture. And that's why machine learning is important at the beginning.

72
0:19:6-->0:19:21
soWe'll review in the next class machine learning basics, and then we'll jump into nlp and language representations. By the way, i'm going to be following the book, outlined in our silver stated in our syllabus.

73
0:19:22-->0:19:38
umBelieve it's literally just called natural language processing. umThe chapters all out are generally outlined here. But in each lecture, i'll state the specific sections I want you to

focus on.

74
0:19:38-->0:19:41
It'll be most of the chapter, but all amid certain portions.

75
0:19:41-->0:19:55
soAt first, we're going to go over certain topics in chapter two. Sorry. First, we're going to go over machine learning, and then we'll go over to ah certain topics in chapter two.

76
0:19:57-->0:19:58
Excuse me,

77
0:19:59-->0:20:20
andthis will largely be about what how how language itself is represented. soHere we'll go over in chapter two, for example, what is a word? How do we process a word? How do we identify a word?

78
0:20:21-->0:20:37
Again, this is all going to be text based. So um we're going to understand what a word in the sense of a textual word is, not the audio word that i'm talking to you using. So

79
0:20:37-->0:20:52
this will include things like reg xa regular expression, which is something you can use to capture strings of symbols, as well as other concepts.

80
0:20:52-->0:20:59
soThis will be about how we represent natural language and capture it.

81
0:20:59-->0:21:18
In just plain text. What? in other wordsIn other words, what is a word? What are things like words? And how do we begin to process them next in chapter? ohAnd by the way, as a heads up, this will be will mostly be going through chapter.

82
0:21:18-->0:21:40
I can tell you this now 2.12.2, and then 2.4~2.7. And then also 2.9.

83
0:21:42-->0:22:1
ah you knowWhat i'll go through the chapters that explicitly to go through here. Next we're going to go over language models. A language model, as we'll discuss in greater detail, then, is a statistical, the key thing here is statistical

84
0:22:1-->0:22:17
model or representation of a language. As we'll see, it's based on the vocabulary that we introduce in chapter two. umThe key chapters here are chapters 3.1, the intro of course read

two,

85
0:22:17-->0:22:44
3.3, and then 3.5 and 3.6. That's for chapter three for three. For chapter four, do 4.1~4.3. Sorry, 4.4.

86
0:22:48-->0:23:9
Yeah, actually, II think ah all of chapter four all of chapter four never mind, actually. So this will be about language models and some easy applications in terms of the classifieds. In it. As we'll see, language models are basically

87
0:23:9-->0:23:33
distributions of sequences of words. From there, eventually we'll get into large language models for regression and vector semantics, all of chapter five and chapter six for neural networks and sequence labeling.

88
0:23:34-->0:23:58
I'll go through this in detail or specifically once we get to it. But most of chapter seven and chapter eight, all of chapter nine for deep learning architectures, all of chapter ten for machine translation. And let me see,

89
0:24:3-->0:24:19
yes allin all of these chapters, the whole chapters, unless I specify, specifically in the course, so okaygoing back into this neural networks and sequence labeling, this isthis will be an application of

90
0:24:21-->0:24:22
language models in a way.

91
0:24:22-->0:24:36
Basically, you're going to use neural networks to construct language. Models, sequence labeling, as we'll see, has to do with how we label sequences of texts or sorry, sequences of words,

92
0:24:36-->0:24:46
in text and some broader text, as either parts of a speech or good or bad are basically characterizing them.

93
0:24:46-->0:24:51
Then we'll dive into a deep learning architectures. right soThis will be fundamental neural networks.

94
0:24:52-->0:24:56
Deep learning will be basically just larger neural networks.

95
0:24:56-->0:25:10
Machine translation here will understand how we go from a text in one language to text in a different language that may or may not even have the same alphabet using this concept of context or context vector.

96
0:25:11-->0:25:29
Transfer learning in constituency grammar is okay, this is interesting. Transfer learning is a bit of its own topic. It's separate than constituency grammar here. We'll introduce in grammar, the concept of a formal or will introduce greater detail, formal

97
0:25:29-->0:25:49
languages and formal grammar here. We'll understand exactly what grammar what grammar really means in both a natural and formal sense. umFrom this, we'll be able to parse parts of a sentence, right? As constituents

98
0:25:50-->0:25:54
like via its constituents and identify its dependencies.

99
0:25:54-->0:26:9
So in other words, as you'll see, we can use the concept of a tree um to represent a sentence and parse that tree where each of the branches are a constituent of a sentence.

100
0:26:11-->0:26:27
After that, we'll switch back into using our more statistical and less concrete, formal methods. uhWhen trying to answer questions, by the way, a lot of this is going to be using what we call beige and statistics. Um

101
0:26:27-->0:26:44
We'll visualize paths for answering questions, and then later in the semester, we'll go into like modern applications, such as chat bots and dialogue systems. Again, all of this will have to do with Beijing statistics, which are enhanced by neural networks as well as

102
0:26:45-->0:26:48
typical NLP techniques, like constituency, independent parsing.

103
0:26:48-->0:27:9
soIt's a bit scattered, but it gives you a broad, enough overview of the techniques we use to process natural language. At the end of the day, the goal of NLP

104
0:27:9-->0:27:30
is to extract the meaning from natural language using these formal methods. So hopefully that gives us an overview. And again, formal methods uh and language are like Python math, machine learning, logic. We'll be using all of this,

105

0:27:31-->0:27:39
all those methods to process natural language and see what meaning we can get out of natural language. That is all, but I will see you next lecture.